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Application of Mathematical Models Based on Differential Equations for Predicting Metformin Dynamics in Patients with Type II Diabetes Mellitus

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Abstract

Type 2 diabetes mellitus (T2DM) represents a major global health challenge, with metformin serving as the first-line pharmacological treatment. This study develops and validates a mathematical model based on ordinary differential equations (ODEs) to predict metformin pharmacokinetics in patients with T2DM. We implemented a one-compartment model with first-order absorption and elimination kinetics, deriving analytical solutions through matrix diagonalization methods. The model's predictions were compared against classical experimental data from Tucker et al. (1981) following a 500 mg oral dose. Our results show that while the model captures the general shape of the concentration-time curve, significant discrepancies were observed in key parameters: the model predicted a C_{max} of 0.98 µg/mL and T_{max} of 4.6 hours, compared to experimental values of 1.12 µg/mL and 2.0 hours, respectively. Sensitivity analysis revealed that the model is particularly sensitive to variations in the absorption rate constant. These findings suggest that the one-compartment model with first-order absorption oversimplifies metformin's pharmacokinetic profile, particularly underestimating the initial absorption rate. Nevertheless, the model provides a valuable foundation for exploring dosing regimens and highlights the need for more complex models that incorporate saturable transport mechanisms to better represent metformin's biological behavior.

Keywords: metformin; pharmacokinetics; ordinary differential equations; mathematical modeling; diabetes mellitus

Introduction

Type 2 diabetes mellitus (T2DM) is a chronic metabolic disorder characterized by persistent hyperglycemia resulting from a combination of insulin resistance and inadequate compensatory insulin secretion by pancreatic beta cells (Hossain *et al.* 2024). This form of diabetes accounts for approximately 90–95% of all diabetes cases worldwide and is closely associated with genetic predisposition, obesity, and sedentary lifestyle habits. In its early stages, T2DM may be asymptomatic or manifest only with mild, non-specific symptoms. When present, typical manifestations include excessive thirst (polydipsia), frequent urination (polyuria), fatigue, blurred vision, and unintentional weight loss. If left uncontrolled, chronic hyperglycemia can lead to serious long-term complications, including microvascular damage (retinopathy, nephropathy, and neuropathy) and macrovascular events such as myocardial infarction and stroke. These complications substantially increase morbidity and mortality and have a profound negative impact on the quality of life of affected individuals. T2DM constitutes a major global public health challenge due to its rising prevalence. According to the World Health Organization, the number of people living with diabetes rose from approximately 200 million in 1990 to more than 830 million in 2022, with the vast majority of these cases corresponding to type 2 diabetes (World Health Organization 2024). This increase is particularly pronounced in low- and middle-income countries. Moreover, T2DM is one of the leading causes of blindness, kidney failure, cardiovascular disease, stroke, and lower-limb amputation worldwide (World Health Organization 2024). Together, these observations underscore the substantial health and economic burden imposed by T2DM at both the individual and societal levels.

Metformin Pharmacological Overview

Metformin is a synthetic *biguanide* oral antihyperglycemic agent widely used as the first-line pharmacotherapy for type 2 diabetes mellitus (T2DM) (Froldi *et al.* 2024; Zhu *et al.* 2023). Its glucose-lowering effect is primarily achieved by suppressing hepatic gluconeogenesis through inhibition of mitochondrial respiratory-chain complex I, which increases the cellular AMP/ATP ratio and activates AMP-activated protein kinase (AMPK) (Froldi *et al.* 2024). In addition, metformin improves peripheral insulin sensitivity and reduces intestinal glucose absorption; it also stimulates glucagon-like peptide-1 (GLP-1) secretion and modulates the gut microbiota, all of which contribute to glycemic control and other metabolic benefits (Cheng *et al.* 2024). Pharmacokinetic studies indicate that metformin has moderate oral bioavailability, which decreases when taken with food, and that it is extensively distributed into tissues (large apparent volume of distribution) while binding minimally to plasma proteins (Froldi *et al.* 2024). Notably, metformin undergoes no significant hepatic metabolism and is excreted largely unchanged by the kidneys via active tubular secretion, with an elimination half-life of approximately 5 hours in patients with normal renal function (Froldi *et al.* 2024). Given its central role in T2DM management and widespread clinical use, continued research into metformin's pharmacokinetic and pharmacodynamic behavior remains highly relevant (Cheng *et al.* 2024; Froldi *et al.* 2024). Inter-individual variability in metformin response is well recognized; for example, genetic polymorphisms in organic cation transporter genes have been shown to affect metformin absorption and elimination, leading to differences in glycemic efficacy between patients (Pérez-Gómez *et al.* 2023). Understanding these sources of variability is critical for optimizing dosing regimens and personalizing therapy, as proper pharmacokinetic profiling can help maximize therapeutic benefit while minimizing adverse effects. In T2DM patients with comorbid conditions such as renal impairment, reduced drug clearance can elevate metformin levels and heighten the risk of lactic acidosis, underscoring the need for careful PK monitoring and dose adjustment (Froldi *et al.* 2024). Furthermore, ongoing studies continue to uncover novel mechanisms and pleiotropic effects of metformin, ranging from anti-inflammatory to potential anti-aging actions, highlighting the value of deeper insight into its dynamic behavior to guide safer and more effective use in clinical practice (Zhu *et al.* 2023).

Ordinary Differential Equations (ODEs)

Mathematically, an ordinary differential equation (ODE) is an equation of the form

$$F\left(t, x, \frac{dx}{dt}, \frac{d^2x}{dt^2}, \ldots, \frac{d^nx}{dt^n}\right) = 0 \tag{1}$$

where $x : \mathbb{R} \to \mathbb{R}^n$ is an unknown function of the variable t and d^kx/dt^k denotes the k-th derivative of $x(t)$ with respect to t. The **order** of an ODE is defined as the highest order of derivative that appears in (1). For instance, an equation involving only the first derivative is called a *first-order ordinary differential equation*, and we shall focus our attention on equations of this type. In that case, equation (1) reduces to

$$F\left(t, x, \frac{dx}{dt}\right) = 0 \tag{2}$$

An equivalent formulation is the so-called **normal form** of a first-order ODE,

$$\frac{dx}{dt} = f(t, x) \tag{3}$$

where $f : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is a given function (Mitsui and Hu 2023). The solutions of (3) generally form a **family** of functions that satisfy the equality. In order to single out a particular solution, one prescribes an initial condition. The problem then takes the form

$$\frac{dx}{dt} = f(t, x) \qquad t_0 < t < t_F \qquad x(t_0) = \xi \tag{4}$$

where $t_0, t_F \in \mathbb{R}$ and $\xi \in \mathbb{R}^n$. This is called an initial-value problem. Under suitable regularity assumptions on f (for instance, continuity and local Lipschitz continuity in the variable x), one can guarantee **existence and uniqueness** of solutions to this problem (Mitsui and Hu 2023). Such initial-value problems arise in the mathematical modeling of many deterministic processes, such as those encountered in physics, engineering, and, of particular interest here, pharmacokinetics.

Pharmacokinetics

Pharmacokinetics (PK) is the branch of pharmacology dedicated to quantifying *what the body does to a drug*, encompassing its Absorption, Distribution, Metabolism, and Excretion (ADME) (Riviere 2011). It is formally defined as the use of mathematical models to characterize the time course of drug concentrations in the body. Because PK processes are dynamic and time-dependent, their behavior is naturally described by differential equations (Riviere 2011; Scheen 1996). By formulating differential equations for ADME processes, one can predict how quickly a drug is absorbed into systemic circulation, how it distributes and accumulates, and how it is eliminated over time. These models allow simulation of concentration–time profiles under various physiological conditions, making PK an essential tool for dose optimization and for understanding drug behavior *in vivo*. Within this framework, we employ a one-compartment pharmacokinetic model with first-order absorption and elimination to describe the kinetics of metformin. This model assumes that the body can be represented as a single, well-mixed compartment where drug distribution is instantaneous. After oral administration, metformin enters the compartment from the gastrointestinal tract at a rate proportional to the remaining unabsorbed drug, governed by the first-order absorption constant k_a. It is simultaneously eliminated from the compartment at a rate proportional to the amount of drug present, described by the first-order elimination constant k_e. The model is expressed as the following system of ordinary differential equations:

$$\begin{aligned} \frac{dA_g}{dt} &= -k_a A_g \\ \frac{dA_c}{dt} &= k_a A_g - k_e A_c \end{aligned} \tag{5}$$

where $A_g(t)$ denotes the amount of drug in the gastrointestinal tract and $A_c(t)$ the amount in the central compartment (systemic circulation). The model assumes linear kinetics, implying that the fraction of drug absorbed or eliminated per unit time remains constant regardless of the dose. This system naturally gives rise to an initial value problem once initial conditions are specified. For a single oral dose D_0 administered at time $t_0 = 0$, the gastrointestinal compartment initially contains the entire dose while the central compartment contains none. Thus, the system becomes

$$\begin{cases} \dfrac{dA_g}{dt} = -k_a A_g \\ \dfrac{dA_c}{dt} = k_a A_g - k_e A_c \end{cases} \qquad A_g(0) = D_0, \quad A_c(0) = 0. \tag{6}$$

This formulation as a P.V.I. (initial value problem) allows one to determine a unique solution under standard regularity assumptions, enabling numerical simulation of the concentration–time profiles associated with a given dose. This simplified representation is particularly suitable for metformin due to several pharmacokinetic characteristics. Metformin is absorbed primarily in the small intestine and, although its uptake involves organic cation transporters and can be dose-dependent at high concentrations, within therapeutic ranges it behaves approximately as a first-order process (Scheen 1996). Once absorbed, it distributes rapidly with minimal plasma protein binding and is not significantly metabolized by the liver. Its elimination occurs predominantly through renal excretion in unchanged form, with a relatively constant clearance and a dose-independent half-life of 4–8 hours in individuals with normal renal function (Scheen

1996). These properties make metformin an ideal candidate for modeling with first-order kinetics in a single compartment.

Materials and methods

Existence and uniqueness of solutions

The mathematical model described in (6) and employed in this work is a special case of the initial value problem introduced in Definition 4. Therefore, in order to justify the existence and uniqueness of solutions of the i.v.p. (6), we resort to a classical result from the theory of ordinary differential equations.

Theorem 1 (Picard–Lindelöf)**.** *Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be a domain and suppose that $f : D \to \mathbb{R}$ is continuous on D and satisfies a Lipschitz condition in its second variable; that is, there exists $L > 0$ such that*

$$||f(t, x_1) - f(t, x_2)|| \leq L||x_1 - x_2|| \quad \forall (t, x_1), (t, x_2) \in D.$$

Then, for any point $(t_0, x_0) \in D$, there exists an interval I around t_0 in which the initial value problem

$$\frac{dx}{dt} = f(t, x), \qquad x(t_0) = x_0$$

has a unique solution (Mitsui and Hu 2023).

The previous theorem requires an additional specification of the notation $|| \cdot ||$, since an appropriate norm is needed for the analysis of the system.

Definition 1 (Norm)**.** Let V be a real vector space. A function $\| \cdot \| : V \to [0, \infty)$ is called a norm if, for all $u, v \in V$ and every scalar $\alpha \in \mathbb{R}$, the following properties hold:

1. **Positivity:** $\|u\| \geq 0$ and $\|u\| = 0$ if and only if $u = 0$.
2. **Homogeneity:** $\|\alpha u\| = |\alpha| \, \|u\|$.
3. **Triangle inequality:** $\|u + v\| \leq \|u\| + \|v\|$.

A pair $(V, \| \cdot \|)$ is then called a normed vector space (Axler 2024).

In this study, we consider two canonical norms:

- The *Euclidean norm* on $\mathbb{R}^n$, defined by

$$\|x\|_2 = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}$$

- The *induced norm* on the matrix space $\mathbb{R}^{n \times n}$, defined in terms of the Euclidean norm, for $x \in \mathbb{R}^n$,

$$\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{||x||_2}$$

These norms are used throughout the analysis to quantify the magnitude of vectors and the amplification properties of linear operators arising in the dynamical model. In particular, both norms are required to obtain quantitative information about the system.

Having guaranteed existence and uniqueness of solutions to (6), the next step is to identify an appropriate method to solve it.

Matrix diagonalization method

The solution method that will be used to study the problem is based on the following theorems.

Theorem 2 (Linear systems diagonalization)**.** *Consider the system of linear ordinary differential equations:*

$$\frac{dx}{dt} = Ax$$

where $\mathbf{x} \in \mathbb{R}^n$ and A is an $n \times n$ diagonalizable matrix.

If $A = PDP^{-1}$, where:

- *$D = diag(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is the diagonal matrix of eigenvalues,*
- *$P = (\mathbf{v}_1 | \mathbf{v}_2 | \cdots | \mathbf{v}_n)$ is the matrix of eigenvectors,*

then the general solution of the system is

$$\mathbf{x}(t) = e^{At} x(t_0) = P e^{Dt} P^{-1} x(t_0),$$

where $e^{Dt} = diag(e^{\lambda_1 t}, e^{\lambda_2 t}, \ldots, e^{\lambda_n t})$.

The next result allows us to simplify the system of differential equations by transforming it into a set of independent scalar equations.

Theorem 3 (Decoupling theorem)**.** *The change of variables $\mathbf{y} = P^{-1}\mathbf{x}$ transforms the coupled system*

$$\frac{dx}{dt} = Ax$$

into the decoupled system

$$\frac{dy}{dt} = By,$$

whose equations are independent:

$$\frac{dy_i}{dt} = \lambda_i y_i, \quad i = 1, 2, \ldots, n,$$

with elementary solutions $y_i(t) = y_i(0) e^{\lambda_i t}$.

Independency of solutions

Since this method is based on Linear Algebra techniques, indeed, this method finds a base of the solution space of the differential equation 5, we need to verify that these solutions are linearly independent. Hence, we need the following definition.

Definition 2. Let $\mathscr{C}^1(I)$ denote the space of continous functions with continous derrivative on an interval $I \subseteq \mathbb{R}$. The **Wronskian** of $f, g \in \mathscr{C}^1(I)$ is defined as:

$$W(f,g)(x) = \begin{vmatrix} f(x) & g(x) \\ f'(x) & g'(x) \end{vmatrix} = f(x) \cdot g(x) - g(x) \cdot f(x)$$

More specifically, we need the followoing result, who might help us determining the solutions of 5

Theorem 4 (Linear Independency)**.** *Let f and g be two differentiable functions on an interval I. If there exists $x_0 \in I$ such that*

$$W(f,g)(x_0) = f(x_0)g'(x_0) - f'(x_0)g(x_0) \neq 0,$$

then f and g are linearly independent on I.

All numerical computations and graphical representations were carried out using the programming language `Python` (version 3.14). Linear algebra operations, evaluation of the analytical solution, and manipulation of parameter values were implemented with the `NumPy` library, which provides efficient vectorized routines on $\mathbb{R}^n$ and $\mathbb{R}^{n\times n}$. The trajectories of the solutions and phase plots associated with system (6) and its equivalent formulation (7) were visualized using the `Matplotlib` library.

All scripts were executed on a standard desktop environment, and the numerical results reported in Section can be fully reproduced from the corresponding `Python` code. The implementation strictly follows the analytical expressions derived in the previous subsections, ensuring consistency between the theoretical results and their numerical counterparts.

In the interest of transparency, we explicitly acknowledge the use of an artificial intelligence tool (ChatGPT, OpenAI) as a writing assistant to improve the clarity, fluency, and consistency of the English text in this manuscript. The mathematical formulation of the model, the derivation of the equations, the proofs, and the interpretation of the results were entirely conceived, verified, and are fully endorsed by the authors. No generative AI tool was used to design the model,

Model Validation with Experimental Data

The predictions of the pharmacokinetic model were validated against experimental data from the study by **?** for the following reasons:

1. **Standard Reference:** This work is considered a classic and widely cited study in the pharmacokinetics of metformin, making it a valid and accepted benchmark for comparison.

2. **Rigorous Methodology:** The authors administered a single 500 mg oral dose of metformin and quantified plasma concentrations using precise analytical methods (GC-MS), ensuring data reliability.

3. **Key Reported Parameters:** The study explicitly reports the essential parameters for our validation:

The use of this high-quality dataset enables a robust evaluation of our model's predictive capability against the real pharmacokinetic behavior of metformin.

Results

Recall our main object of study: we consider the autonomous linear system with constant coefficients

$$\begin{cases} \dfrac{dA_g}{dt} = -k_a A_g \\ \dfrac{dA_c}{dt} = k_a A_g - k_e A_c \end{cases} \qquad A_g(0) = D_0, \quad A_c(0) = 0.$$

where $k_a > 0$ and $k_e > 0$. This system corresponds to the particular instance of the general model introduced in (3) and (6).

We now rewrite the system in compact form. Let us define the state vector x as

$$x = \begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix} \in \mathbb{R}^2$$

and the matrix

$$A = \begin{bmatrix} -k_a & 0 \\ k_a & -k_e \end{bmatrix}.$$

Then, problem 6 becomes

$$\frac{dx}{dt} = Ax \qquad x(0) = \begin{pmatrix} D_0 \\ 0 \end{pmatrix} \tag{7}$$

where $f(t,x) = f(x) = Ax$. This formulation is equivalent to the original problem and will be used to simplify the subsequent analysis.

Proposition. The i.v.p. 6 has a unique solution for all $x \in \mathbb{R}^2$.

Proof. We equip $\mathbb{R}^2$ with the Euclidean norm $\|\cdot\|$. Since A is a fixed 2×2 matrix, its induced matrix norm

$$\|A\| := \sup_{z\neq 0} \frac{\|Az\|}{\|z\|}$$

is finite. Hence, for every $z \in \mathbb{R}^2$ we have the estimate

$$\|Az\| \le \|A\|\,\|z\|.$$

Let $x, y \in \mathbb{R}^2$ be arbitrary. Using the linearity of A we obtain

$$f(x) - f(y) = Ax - Ay = A(x-y),$$

and therefore

$$\|f(x) - f(y)\| = \|A(x-y)\| \le \|A\|\,\|x-y\|.$$

This shows that f is globally Lipschitz continuous on $\mathbb{R}^2$, with Lipschitz constant $L = \|A\|$. In particular, every Lipschitz function is continuous, so f is continuous at every point of $\mathbb{R}^2$. ■

Having guaranteed the existence and uniqueness of the solution to (7), we are now in a position to solve the i.v.p. explicitly by first determining its fundamental solutions using the diagonalization method described in Section .

Analytical solution

Let us recall equation 7:

$$\frac{d}{dt}\begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix} = \begin{bmatrix} -k_a & 0 \\ k_a & -k_e \end{bmatrix} \begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix}. \tag{8}$$

We first compute the characteristic polynomial by evaluating the determinant

$$\det(A - \lambda I) = \det\left(\begin{bmatrix} -k_a & 0 \\ k_a & -k_e \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right).$$

Since A is a lower triangular matrix, this determinant is easily obtained as the product of the entries on the main diagonal, and the characteristic polynomial is given by

$$p(\lambda) = (-k_a - \lambda)\cdot(-k_e - \lambda).$$

In order to obtain the eigenvalues, we find the zeros of the characteristic polynomial:

$$(k_a + \lambda)(k_e + \lambda) = 0$$
$$\lambda_1 = -k_a, \quad \lambda_2 = -k_e.$$

The next step is to determine the eigenvectors associated with each λ_1 and λ_2 by solving the linear system $(A-\lambda_i I)v_i = 0_{\mathbb{R}^2}$, where $v_i \in \mathbb{R}^2$.

For $\lambda_1 = -k_a$:

$$(A-\lambda_1 I)\mathbf{v}_1 = \begin{pmatrix} 0 & 0 \\ k_a & -k_e+k_a \end{pmatrix}\begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

We choose $\mathbf{v}_1 = \begin{pmatrix} k_e - k_a \\ k_a \end{pmatrix}$.

For $\lambda_2 = -k_e$:

$$(M-\lambda_2 I)\mathbf{v}_2 = \begin{pmatrix} -k_a+k_e & 0 \\ k_a & 0 \end{pmatrix}\begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

We choose $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

We are now in a position to define the matrix P as in Theorem 2 and to compute its inverse as follows:

$$P = \begin{pmatrix} k_e-k_a & 0 \\ k_a & 1 \end{pmatrix}, \quad P^{-1} = \frac{1}{k_e-k_a}\begin{pmatrix} 1 & 0 \\ -k_a & k_e-k_a \end{pmatrix},$$

in order to apply Theorem 3 and transform (8) into a simpler system by means of the change of variables $y = P^{-1}x$, i.e.,

$$y = P^{-1}x = \begin{pmatrix} 1 & 0 \\ -k_a & k_e-k_a \end{pmatrix}\begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix}. \tag{9}$$

Of course, we can recover the original variables through the transformation $x = Py$, i.e.,

$$x = Py = \begin{pmatrix} k_e-k_a & 0 \\ k_a & 1 \end{pmatrix}\begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix}. \tag{10}$$

Substituting the transformation 10 into the original system, we obtain

$$\frac{dy}{dt} = A(Py),$$

$$P\frac{dy}{dt} = APy,$$

$$\frac{dy}{dt} = P^{-1}APy.$$

But $P^{-1}MP = D = diag(-k_a, -k_e)$, so equation 8 transforms into

$$\frac{dy}{dt} = Dy = \begin{pmatrix} -k_a & 0 \\ 0 & -k_e \end{pmatrix} y,$$

which can be written as

$$\begin{cases} \dfrac{dy_1}{dt} = -k_a y_1 \\ \dfrac{dy_2}{dt} = -k_e y_2 \end{cases}$$

and therefore the two equations are now independent. Moreover, each of them corresponds to a first-order linear decay model, whose solutions are well known:

$$y_1(t) = C_1 e^{-k_a t}, \quad y_2(t) = C_2 e^{-k_e t},$$

where $C_1, C_2 \in \mathbb{R}$.

Linear Indepency of the solutions

What we´ve done untl now mathematically is to find a basis of the solution space of the differential equation, but this can only happen when the base is linearly independent, hence we use the Wrosnkian to prove it's independendy,

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} \\ &= \begin{vmatrix} C_1 e^{-k_a t} & C_2 e^{-k_e t} \\ -C_1 k_a e^{-k_a t} & -C_2 k_e e^{-k_e t} \end{vmatrix} \\ &= -\big(C_1 \cdot C_2 \cdot k_e\big) e^{-(k_a t + k_e t)} \\ &\quad - \big(C_1 \cdot C_2 \cdot k_a\big) e^{-(k_a t + k_e t)} \\ &= -\big(C_1 \cdot C_2\big) \cdot \big(k_e e^{-(k_a+k_e)\cdot t} + k_a e^{-(k_a+k_e)\cdot t}\big) \end{aligned}$$

But,

$$\begin{aligned} & e^{(k_a+k_e)t} > 0 \quad \forall t \in \mathbb{R} \\ \implies & \frac{1}{e^{(k_a+k_e)t}} > 0 \\ \implies & e^{-(k_a+k_e)t} > 0 \\ \implies & k_e e^{-(k_a+k_e)t} > 0 \quad \forall t \in \mathbb{R} \end{aligned}$$

Similar analysis can be done to prove $k_a e^{-(k_a+k_e)} > 0$.

Thus,

$$k_e e^{-(k_a+k_e)\cdot t} + k_a e^{-(k_a+k_e)\cdot t} > 0$$

Hence, we can derive the following cases

- $C_1 \cdot C_2 < 0$
  $$W(y_1,y_2)(t) = -\big(C_1 \cdot C_2\big) \cdot \big(k_e e^{-(k_a+k_e)\cdot t} + k_a e^{-(k_a+k_e)\cdot t}\big) > 0$$
- $C_1 \cdot C_2 < 0$
  $$W(y_1,y_2)(t) = -\big(C_1 \cdot C_2\big) \cdot \big(k_e e^{-(k_a+k_e)\cdot t} + k_a e^{-(k_a+k_e)\cdot t}\big) > 0$$
- $C_1 \cdot C_2 = 0$
  $$W(y_1,y_2)(t) = -\big(C_1 \cdot C_2\big) \cdot \big(k_e e^{-(k_a+k_e)\cdot t} + k_a e^{-(k_a+k_e)\cdot t}\big) = 0$$

The previous result whows that the solutions are lenearly independent if and only if

$$C_1 \cdot C_2 \neq 0$$

, because, it´s the only posible situation when the determinant becomes 0.

We now use the initial information about the process, namely the initial conditions, to determine C_1 and C_2. The initial conditions are $A_g(0) = D_0$, $A_c(0) = 0$, so

$$x(0) = \begin{pmatrix} D_0 \\ 0 \end{pmatrix}.$$

Applying the transformation 9 to the initial condition, we obtain

$$y(0) = \frac{1}{k_e-k_a}\begin{pmatrix} 1 & 0 \\ -k_a & k_e-k_a \end{pmatrix}\begin{pmatrix} D_0 \\ 0 \end{pmatrix} = \begin{pmatrix} \dfrac{D_0}{k_e-k_a} \\ -\dfrac{k_a D_0}{k_e-k_a} \end{pmatrix}.$$

Thus, the solution in terms of the variable y is given by

$$y_1(t) = \frac{A_0}{k_e - k_a} e^{-k_a t}, \quad y_2(t) = -\frac{k_a A_0}{k_e - k_a} e^{-k_e t}.$$

We still need to express the solution in terms of the original variable x via the transformation 10:

$$\begin{pmatrix} A_g(t) \\ A_c(t) \end{pmatrix} = \begin{pmatrix} k_e - k_a & 0 \\ k_a & 1 \end{pmatrix} \begin{pmatrix} \dfrac{A_0}{k_e - k_a} e^{-k_a t} \\ -\dfrac{k_a A_0}{k_e - k_a} e^{-k_e t} \end{pmatrix}.$$

Performing the matrix–vector multiplication yields

$$A_g(t) = (k_e - k_a)\left(\frac{A_0}{k_e - k_a} e^{-k_a t}\right) + 0 = A_0 e^{-k_a t},$$

$$\begin{aligned} A_c(t) &= k_a \left(\frac{D_0}{k_e - k_a} e^{-k_a t}\right) + 1\left(-\frac{k_a D_0}{k_e - k_a} e^{-k_e t}\right) \\ &= \frac{k_a D_0}{k_e - k_a}(e^{-k_a t} - e^{-k_e t}). \end{aligned}$$

We thus obtain the final explicit solution

$$\boxed{A_g(t) = D_0 e^{-k_a t}}$$

$$\boxed{A_c(t) = \frac{k_a D_0}{k_e - k_a}(e^{-k_a t} - e^{-k_e t})}.$$

or equivalently:

$$x(t) = \begin{pmatrix} D_0 e^{-k_a t} \\ \frac{k_a D_0}{k_e - k_a}(e^{-k_a t} - e^{-k_e t}) \end{pmatrix}$$

Computational Implementation

All numerical computations and graphical representations were carried out using the programming language `Python` (version 3.14). Linear algebra operations, evaluation of the analytical solution, and manipulation of parameter values were implemented with the `NumPy` library, which provides efficient vectorized routines on $\mathbb{R}^n$ and $\mathbb{R}^{n \times n}$. The trajectories of the solutions and phase plots associated with system (6) and its equivalent formulation (7) were visualized using the `Matplotlib` library.

The complete interactive implementation, including all simulations and analyses presented in this work, is available as a Google Colab notebook:

https://colab.research.google.com/drive/1QrX2CFh4_Sno8pK29T61cUP5URCODC-M?usp=sharing

All scripts were executed on a standard desktop environment, and the numerical results reported in Section can be fully reproduced from the corresponding `Python` code. The implementation strictly follows the analytical expressions derived in the previous subsections, ensuring consistency between the theoretical results and their numerical counterparts.

Pharmacokinetic Simulation and Model Validation

The one-compartment model with first-order absorption was computationally implemented, and its predictions were validated against experimental data reported by Tucker et al. (1981) for a 500 mg oral dose of metformin.

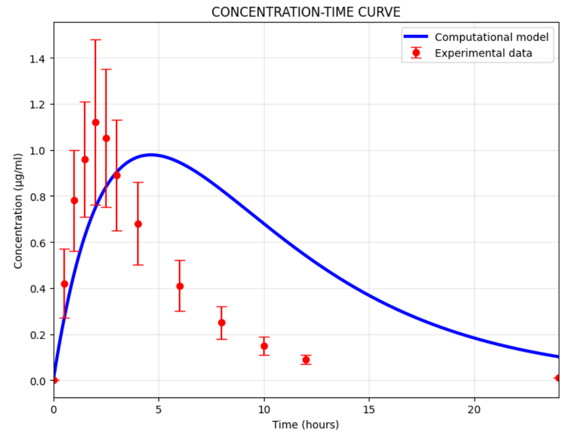

Figure 1 Comparison between the predictions of the one-compartment model and the experimental data reported by Tucker et al., 1981.

Concentration-Time Curve and Model Fit

The simulation reproduced the characteristic shape of the plasma concentration-time curve, showing a rapid increase followed by an exponential decline phase (Figure 1). However, significant quantitative discrepancies were observed between the model predictions and the experimental data. The model predicted a **Cmax of 0.98 µg/ml** compared to the **experimental Cmax of 1.12 µg/ml**, and a **Tmax of 4.6 hours** significantly longer than the **experimental Tmax of 2.0 hours**. This difference suggests that the model underestimates the initial drug absorption rate.

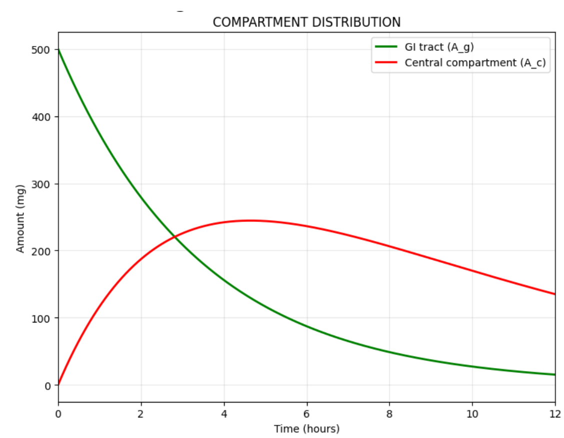

Figure 2 Temporal distribution of the drug amount in the gastrointestinal tract and in the central compartment after oral administration of 500 mg of metformin.

Compartment Distribution and Absorption-Elimination Kinetics

Analysis of drug distribution between the gastrointestinal tract and central compartment revealed that absorption is completed within approximately the first 8 hours (Figure 2). The absorption and elimination rates showed characteristic temporal profiles, with the absorption rate peaking immediately after administration and decaying exponentially, while the elimination rate peaked around 3 hours (Figure 3). Cumulative absorption indicated that 50% of the dose is absorbed in approximately 2.4

hours and 90% in approximately 8 hours (Figure 4).

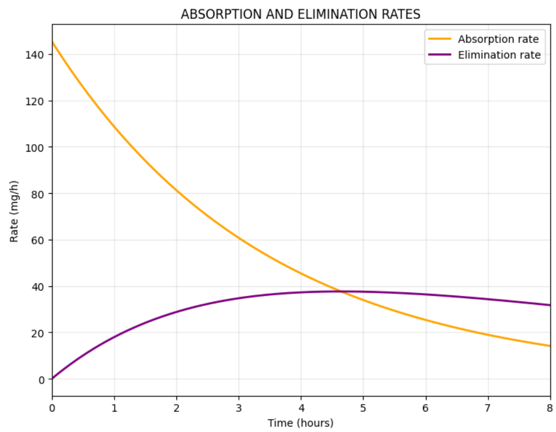

Figure 3 Temporal profiles of the absorption and elimination rates governing the pharmacokinetics of metformin in the one-compartment model.

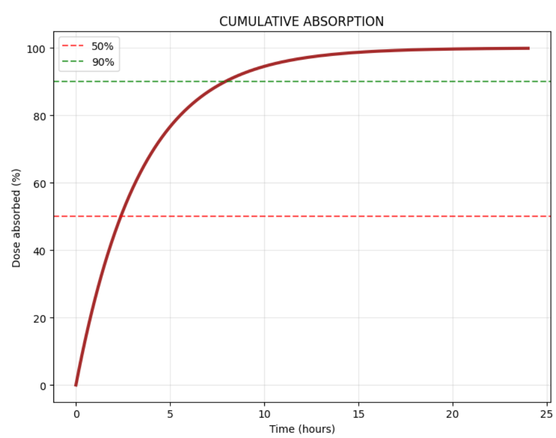

Figure 4 Cumulative percentage of the absorbed dose as a function of time.

Table 1 Pharmacokinetic parameters from model simulation and experimental data

Parameter	Simulated	Experimental
C_{max} (μg/ml)	0.98	1.12
T_{max} (h)	4.6	2.0
AUC (0-24h) (μg·h/ml)	12.32	–
Absorption half-life (h)	2.4	–
Elimination half-life (h)	4.5	–
Estimated bioavailability (%)	615.8	–

Quantitative Evaluation of Model Fit

The quantitative comparison between model predictions and experimental data showed an **average absolute error of 0.344 µg/ml** and an **average relative error of 55.6%** (Table 2). The calculated area under the curve (AUC) was **12.32 µg·h/ml**, although the estimated bioavailability of 615.8% indicates potential limitations in the volume of distribution estimation or model assumptions.

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Table 2 Quantitative comparison between model predictions and experimental data

Time (h)	Experimental (μg/ml)	Model (μg/ml)	Error (μg/ml)	Error (%)
0.5	0.42	0.26	0.16	38.0
1.0	0.78	0.47	0.31	40.2
1.5	0.96	0.63	0.33	34.8
2.0	1.12	0.75	0.37	33.2
2.5	1.05	0.84	0.21	20.2
3.0	0.89	0.90	0.01	1.3
4.0	0.68	0.97	0.29	42.4
6.0	0.41	0.95	0.54	130.5
8.0	0.25	0.83	0.58	230.0
10.0	0.15	0.68	0.53	352.9
12.0	0.09	0.54	0.45	500.0

Multiple Dose Simulation and Sensitivity Analysis

Simulation of different dosing regimens (250, 500, 750 and 1000 mg) demonstrated a linear relationship between dose and maximum concentration achieved (Cmax), with values of 0.49, 0.98, 1.47 and 1.96 µg/ml, respectively (Figure 5).

Sensitivity analysis revealed that the model is particularly sensitive to variations in the absorption rate constant (k_a), where 30% increases advanced and heightened the concentration peak, while 30% decreases significantly delayed and reduced plasma concentrations (Figure 6). Sensitivity to the elimination rate constant (k_e) was less pronounced but showed expected effects on the slope of the decline phase (Figure 7).

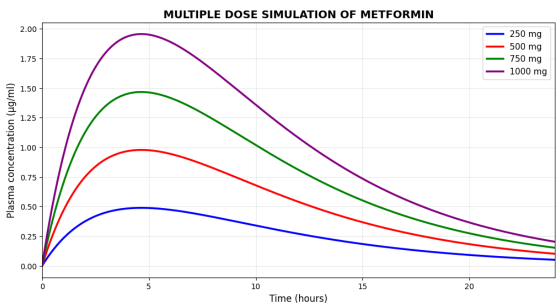

Figure 5 Simulation of concentration-time curves for different oral doses of metformin, illustrating the dose-response relationship of the model.

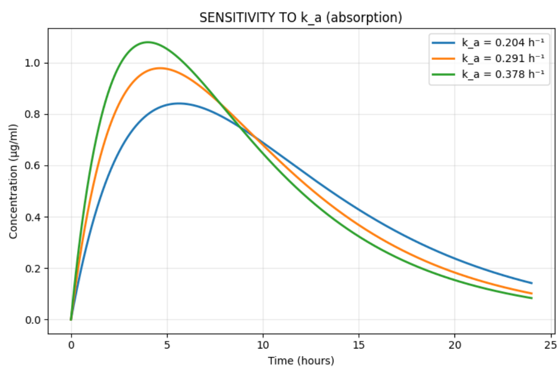

Figure 6 Sensitivity analysis of plasma concentration to $\pm 30\%$ variations in the absorption rate constant, showing its impact on $T_{\max}$ and $C_{\max}$.

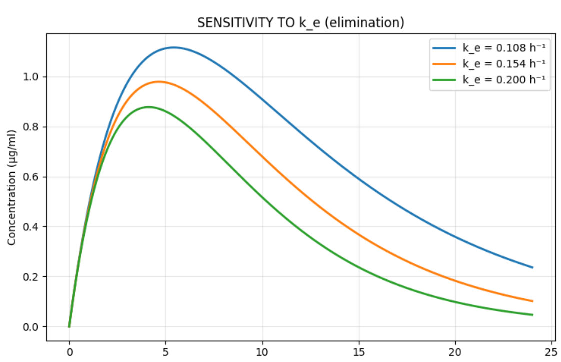

Figure 7 Sensitivity analysis of plasma concentration to $\pm 30\%$ variations in the elimination rate constant, showing its effect on the slope of the decline phase.

Discussion

This study developed a one-compartment pharmacokinetic model with first-order kinetics to describe the metformin profile. While the model captured the general shape of the concentration curve, significant discrepancies were observed with experimental data from Tucker et al. (1981), particularly in the underestimation of T_{max} (4.6 h vs 2.0 h experimental) and overestimation of bioavailability (615.8%). These differences suggest that the assumptions of first-order absorption and instantaneous distribution oversimplify the actual processes of metformin, whose absorption involves saturable transporters (OCTs) and complex tissue distribution.

Sensitivity analysis confirmed the model's high dependence on variations in the absorption constant (k_a), consistent with clinical observations of interindividual variability in metformin response. Despite its limitations, the model demonstrated a linear dose-C_{max} relationship that supports dose titration strategies in clinical practice.

Future Directions

- Implement a bicompartmental model to better capture distribution phases
- Incorporate Michaelis-Menten kinetics for saturable transport processes
- Integrate physiological parameters (renal function, transporter expression)
- Develop population pharmacokinetic models considering genetic variability

In conclusion, while the current model provides a useful foundation, more physiological approaches that incorporate saturable transport mechanisms and multicompartmental distribution are required for accurate prediction of metformin pharmacokinetics.

Code Availability

The complete Python code for this study, implemented as an interactive Google Colab notebook, is available at:

https://colab.research.google.com/drive/1QrX2CFh4_Sno8pK29T61cUP5URCODC-M?usp=sharing

This resource allows readers to reproduce all simulations, modify parameters, and explore the model behavior interactively.

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Conflicts of interest

The authors declare that they have no conflicts of interest relevant to this work. This research was conducted as part of academic activities at UNAM and did not receive any commercial funding.

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